

Truncated Back prop. Through Time ①

Low + BPTT — Const U, V, W .

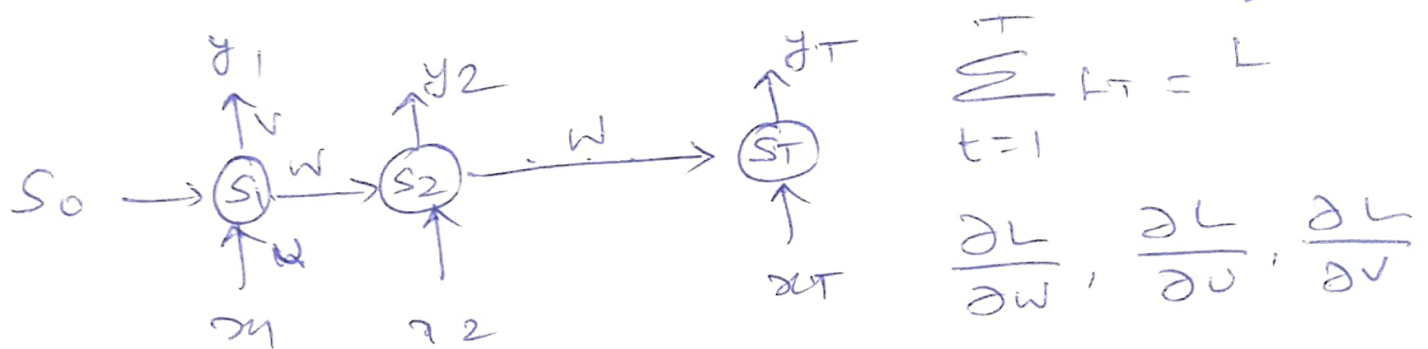
* Gradient calculation either explodes (or) Vanish.

* Gradient calculation are expensive.

Exploding Gradient $\rightarrow$ Gradient clipping.

Vanishing Gradient $\rightarrow$ Alternate architectures (GRU, LSTM)

Expensive $\rightarrow$ Truncated BPTT.



$$\frac{\partial L_3}{\partial W} \rightarrow \frac{\partial s_3}{\partial W} \rightarrow \frac{\partial s_2}{\partial W} \rightarrow \frac{\partial s_1}{\partial W}$$

$$\frac{\partial L_3}{\partial U} \rightarrow \frac{\partial s_3}{\partial U} \rightarrow \frac{\partial s_2}{\partial U} \rightarrow \frac{\partial s_1}{\partial U}$$

$\rightarrow$ dependencies on solving weight ~~update~~ updation.

Let us solve by heuristic description
 Heuristic description $s_t = \phi(Ux_t + Ws_{t-1})$ (or) Rough description
 Rough.

Someone that ignore x_t (It can't be but for experimental) ②

$$s_t \sim \phi(ws_{t-1})$$

Someone that ignore ϕ

$$s_t \sim ws_{t-1}$$

s_t affected by w times of s_{t-1}
Further approximation as

$$s_{t+1} \sim w^2 s_{t-1} [nw_t \sim w^2 t_{t-1}]$$

$$s_{t+n} \sim w^{n+1} s_{t-1} (a)$$

$$s_{t+n} \sim w^n s_t \text{ using this heuristic approx}$$

Let us have a mem of $s_t \sim s_t$ is a vector with two mem.

$$\|A\| = \frac{\|\vec{Av}\|_2}{\|\vec{v}\|_2}$$

$$\|s_{t+n}\| \sim \lambda^n \|s_t\|$$

$\rightarrow$ eigen values of w

If $|\lambda| > 1$ and if 'n' increases

$$\|s_{t+n}\| \rightarrow \infty$$

exploding gradient

(becomes very large)

If $|\lambda| < 1$ and if 'n' increases

$$\|s_{t+n}\| \rightarrow 0.$$

$\downarrow$

Vanishing gradients

$$\frac{\partial L_3}{\partial w} \rightarrow \frac{\partial S_3}{\partial w} \rightarrow w \frac{\partial S_2}{\partial w}$$

$$\left\| \frac{\partial L}{\partial w} \right\| \rightarrow \infty \quad \text{exploding gradient.}$$

$$\left\| \frac{\partial L}{\partial w} \right\| \rightarrow 0 \quad \text{vanishing gradients.}$$

Training very difficult.



Exploding Gradients: $\rightarrow$ Gradient clipping.

$\rightarrow$ Decide on a maximum allowable gradient value/size.

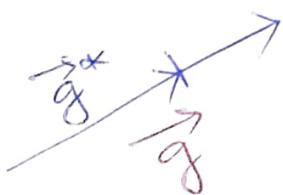
$$\text{Say } \frac{\partial L}{\partial w} = \vec{g} \quad \max \|\vec{g}\| = G_{\max}$$

While Gradient descent

$\rightarrow$ Calculate $\vec{g}$

$\rightarrow$ If $\|\vec{g}\| < G_{\max}$ proceed as usual.

$\rightarrow$ If not, $\vec{g}^* = \frac{\vec{g}}{\|\vec{g}\|} G_{\max}$



$$\vec{g} = \hat{g} G_{\max}$$

Unit Scalar

Vanishing Gradients

(4)

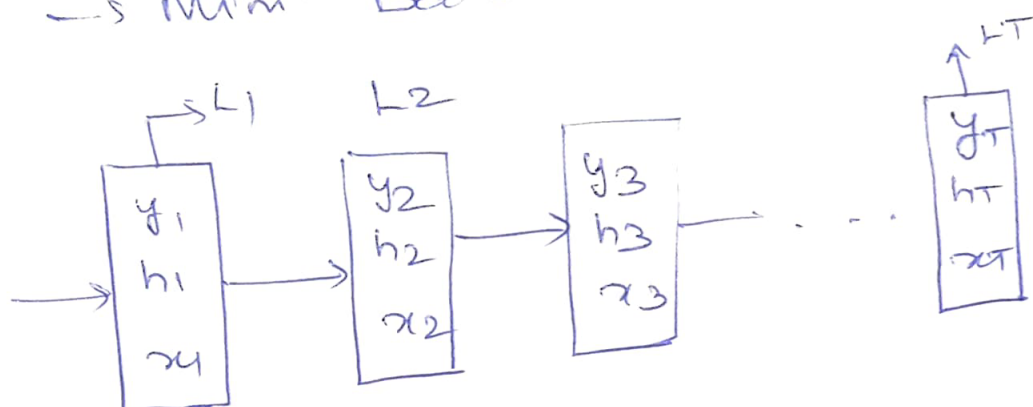
No successful solutions

→ GRU, LSTM.

Computation of gradient is expensive

→ Truncated BPTT.

→ Mini-Batch Gradient descent.



In general,

Forward propagate through K_1 steps

Back propagate through K_2 steps

() your gradient will never blow up
() never vanishes.